joining the tops of such ordinates is known as total energy line (T.E.L).

7.5.4 Loss of head in pipes

When a liquid flows under pressure in a pipe, there will be some losses. These losses are classified as below

1. Major or primary loss (frictional loss)
2. Minor or secondary loss (other losses)

Major loss is mainly due to viscosity, turbulence and roughness of pipe. This loss is called loss due to friction and is denoted by h_f. The losses due to other causes are called minor losses.

1. Major loss

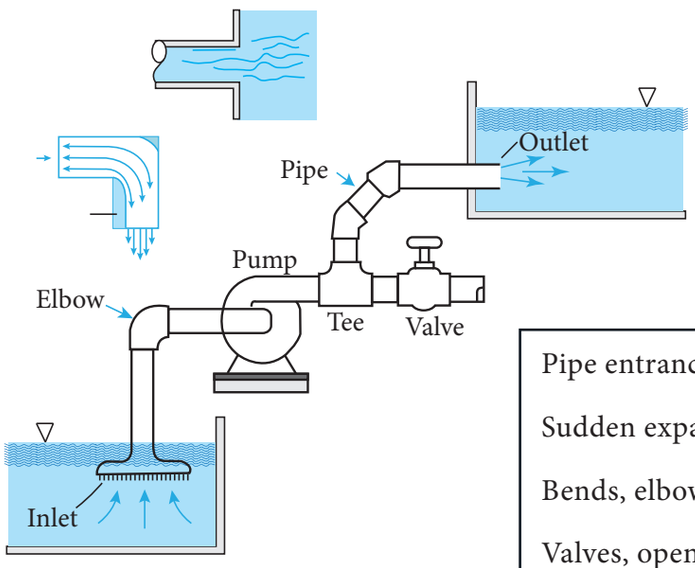
When liquid is flowing in a pipe, it experiences some resistance to its motion due to friction offered by the inner surface of the pipe. This effect is to reduce the velocity and ultimately the head of water. It has been experimentally found that more the roughness of the inside surface of the pipe, greater will be the resistance.

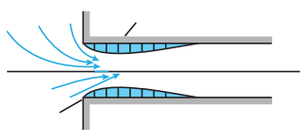
2. Minor losses

The following are the minor losses which occur in a pipe.

- **a.** Loss of head at inlet or entry of a pipe line.
- **b.** Loss of head due to pipe fittings.
- **c.** Loss of head due to change in direction.
- **d.** Loss of head due to sudden enlargement.
- **e.** Loss of head due to sudden contraction.
- **f.** Loss of head due to obstruction.
- **g.** Loss of head at exit of a pipe.

Minor Losses





Pipe entrance or exit

Sudden expansion or contraction

Bends, elbows, tees, and other fittings

Valves, open or partially closed

Gradual expansions or contractions

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